

5. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe`, `dup2()`, and `exec()` system calls.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid;

    // Create pipe
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    // Create child process
    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        exit(1);
    }

    if (pid == 0)
```

```

{
    // Child process: executes grep ".c"

    // Close unused write end
    close(pipefd[1]);

    // Make stdin come from pipe
    dup2(pipefd[0], STDIN_FILENO);

    // Close original read descriptor
    close(pipefd[0]);

    // Execute grep
    execlp("grep", "grep", ".c", NULL);

    // Only reached if exec fails
    perror("execlp");
    exit(1);
}
else
{
    // Parent process: executes ls -l

    // Close unused read end
    close(pipefd[0]);

    // Make stdout go to pipe
    dup2(pipefd[1], STDOUT_FILENO);

    // Close original write descriptor
    close(pipefd[1]);

    // Execute ls -l
    execlp("ls", "ls", "-l", NULL);

    // Only reached if exec fails
    perror("execlp");
    exit(1);
}

return 0;
}

```

Output:

gcc week5.c -o week5

pranav@Pranavs-MacBook-Air dir1 % ./week5

-rw-r--r-- 1 pranav staff 173 27 Jul 14:05 addition.c
-rwxr-xr-x 1 pranav staff 33464 28 Jul 10:33 arithmetic
-rw-r--r-- 1 pranav staff 309 28 Jul 10:32 arithmetic.c
-rwxr-xr-x 1 pranav staff 34080 22 Aug 14:48 command_exec
-rw-r--r-- 1 pranav staff 1317 22 Aug 14:47 command_exec.c
-rwxr-xr-x 1 pranav staff 33641 22 Aug 14:46 execworking
-rw-r--r-- 1 pranav staff 209 22 Aug 14:46 execworking.c
-rw-r--r-- 1 pranav staff 733 4 Aug 10:37 forkid.c
-rw-r--r-- 1 pranav staff 792 4 Aug 10:27 linuxos.c
-rw-r--r-- 1 pranav staff 403 22 Aug 15:08 modzombie1.c
-rwxr-xr-x 1 pranav staff 33569 22 Aug 14:35 multiprocess
-rw-r--r-- 1 pranav staff 165 22 Aug 14:35 multiprocess.c
-rwxr-xr-x 1 pranav staff 33600 11 Aug 10:19 multiprocess2
-rw-r--r-- 1 pranav staff 411 11 Aug 10:19 multiprocess2.c
-rwxr-xr-x 1 pranav staff 33648 11 Aug 10:34 multiprocess3
-rw-r--r-- 1 pranav staff 544 11 Aug 10:33 multiprocess3.c
-rw-r--r-- 1 pranav staff 281 25 Aug 11:13 open.c
-rw-r--r-- 1 pranav staff 578 25 Aug 11:17 open2.c
-rw-r--r-- 1 pranav staff 444 22 Aug 15:14 partb.c
-rw-r--r-- 1 pranav staff 1550 1 Sep 10:22 Priority.c
-rwxr-xr-x 1 pranav staff 33464 30 Jul 11:25 process
-rw-r--r-- 1 pranav staff 191 30 Jul 11:24 process.c
-rwxr-xr-x 1 pranav staff 33464 1 Aug 15:03 process2
-rw-r--r-- 1 pranav staff 245 1 Aug 15:13 process2.c
-rwxr-xr-x 1 pranav staff 33672 1 Aug 15:38 process3
-rw-r--r-- 1 pranav staff 764 1 Aug 15:38 process3.c
-rwxr-xr-x 1 pranav staff 33432 3 Aug 12:26 process4
-rw-r--r-- 1 pranav staff 159 3 Aug 13:31 process4.c
-rwxr-xr-x 1 pranav staff 33672 11 Aug 10:32 process5

```
-rw-r--r-- 1 pranav staff 694 11 Aug 10:33 process5.c
-rw-r--r-- 1 pranav staff 270 25 Aug 11:31 read.c
-rw-r--r-- 1 pranav staff 445 25 Aug 11:39 read2.c
-rw-r--r-- 1 pranav staff 618 3 Sep 10:10 rr.c
-rwxr-xr-x 1 pranav staff 33464 28 Jul 10:53 switchcase
-rw-r--r-- 1 pranav staff 1068 28 Jul 10:53 switchcase.c
-rw-r--r-- 1 pranav staff 1238 3 Sep 10:16 week5.c
-rw-r--r-- 1 pranav staff 208 4 Aug 11:27 write.c
-rw-r--r-- 1 pranav staff 409 11 Aug 10:36 zombie.c
-rw-r--r-- 1 pranav staff 534 22 Aug 15:07 zombie1.c
pranav@Pranavs-MacBook-Air dir1 %
```